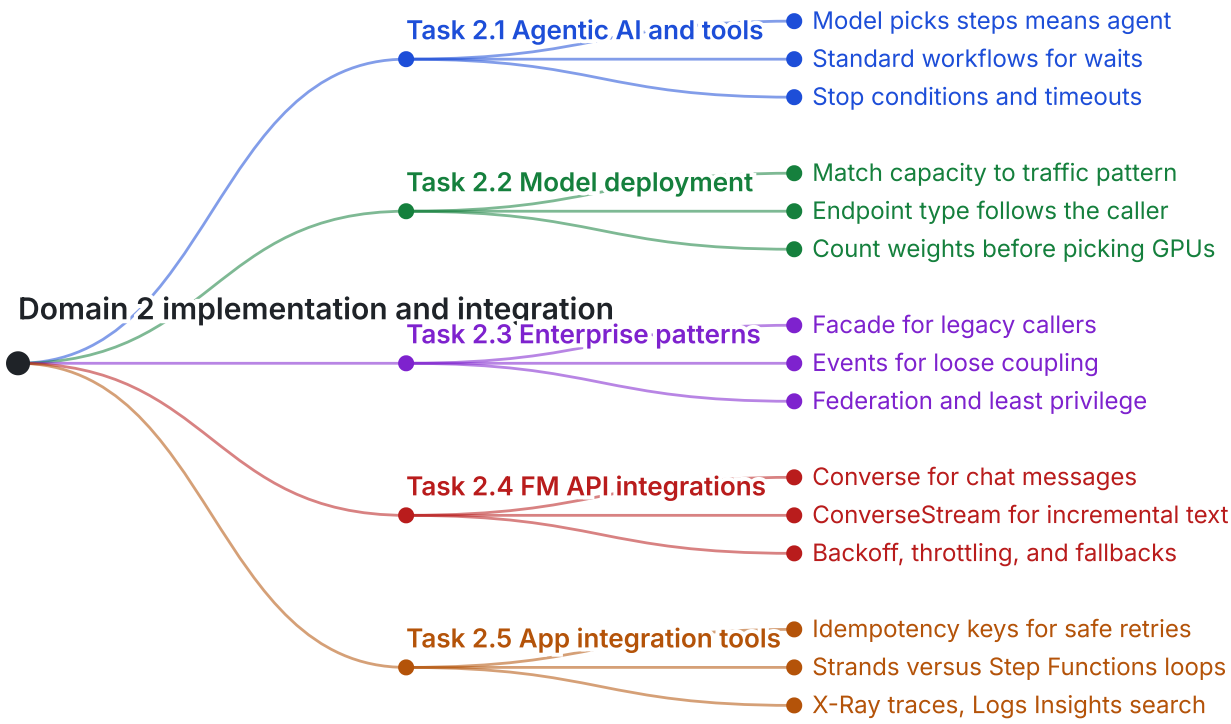
Memory sheet: Domain 2, implementation and integration

Exam weight 26 percent. Read this sheet, then answer the flash questions aloud.



Deciding rules

When the stem says	Think of	Not when
Live, waits for the answer, or as it is generated	Synchronous or streaming call path	No user waits, so a queue or batch job fits
Overnight, backlog, or a job ID	Asynchronous or batch path	The answer must return on the same connection
Per request with no capacity planning	On-demand Bedrock invocation	Steady, high load where reserved capacity costs less
Guaranteed throughput or model units	Provisioned Throughput	Spiky or idle-prone load that pays for unused hours
Large payload kept in S3 with a result read later	Asynchronous endpoint	A live chat that must answer on the same connection
Several models in one family chosen by quality and cost	Intelligent prompt routing	Models span different families, which needs an application router
Streamed reply through API Gateway	REST API with a streaming integration	An HTTP API, which cannot stream responses
A person must approve and the wait lasts days	Standard Step Functions workflow with a callback	Express workflows, which do not support callbacks

Etch lines

1. An HTTP API cannot stream responses, so pick REST for streaming.
2. Throttled calls should retry with backoff and a fixed attempt cap.
3. Asynchronous invocation is not exactly-once, so writes need idempotency keys.
4. Look for live or waiting wording to spot the synchronous path.
5. Batch transform is wrong when a live user waits for each reply.
6. Property checks beat exact string matches for generated output.

Flash questions

- Q: Which API type should carry a streamed chat reply through API Gateway? A: A REST API with a streaming integration.
- Q: Which Step Functions workflow type pauses for a person who may reply days later? A: A Standard workflow with a callback.
- Q: What is the run-time ceiling for one Lambda invocation? A: 15 minutes.
- Q: What stops a retried CRM write from creating a duplicate record? A: An idempotency key based on a stable value, such as an order ID.
- Q: Which routing feature picks among models in one family by quality and cost? A: Intelligent prompt routing.
- Q: Which SageMaker AI endpoint type takes large payloads that need minutes to process? A: An asynchronous endpoint.
- Q: Why does Converse suit a multi-model chat app better than InvokeModel? A: One message shape works across models, with no parser for each model.
- Q: Which private link fits steady, high-volume hybrid traffic that must avoid the internet? A: Direct Connect, with a VPN for failover.

Source: domain2.md, Tasks 2.1 to 2.5, and frags2 t2.1.html to t2.5.html. AWS docs and the study guide sections above, no practice-test text.